CSL7110 Assignment 3 Report

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1. Objective

The objective of this assignment is to implement and compare different recommendation system techniques using the MovieLens dataset.

2. Dataset Used

The MovieLens latest-small dataset was used, which contains movie metadata and user ratings. It is suitable for experimentation and allows efficient implementation in Google Colab.

3. Assumptions

- Ratings greater than or equal to 3 were considered relevant.
- Top-K recommendations were generated for evaluation.
- Due to dataset sparsity, low precision and recall values are expected.
- Genre information was used as the primary feature for content-based filtering.

4. Methods Implemented

The following recommendation techniques were implemented:

- Content-Based Filtering using TF-IDF and cosine similarity
- User Profile-Based Recommendation
- User-Based Collaborative Filtering
- Matrix Factorization using SVD
- Hybrid Recommendation Model

5. Results and Observations

The models were evaluated using Precision@K and Recall@K.

The observed values were relatively low. This is expected because:

- The dataset is sparse
- Basic models were used without hyperparameter tuning
- Exact matching between recommended and relevant items is difficult

Despite this, all models were successfully implemented and produced valid recommendations.

6. GitHub Repository

The complete implementation can be found at:

https://github.com/Yoursuperherobasu/M24AID008_CSL7110_Assignment3.git

7. Conclusion

This assignment helped in understanding different recommendation techniques and their behavior on real-world data. It also highlighted the challenges of evaluation in sparse datasets and the importance of model improvement for better accuracy.

Sample outputs of recommendations and evaluation metrics are included to demonstrate working implementation.

```
Movies: (9742, 3)
Ratings: (100836, 4)
Tags: (3683, 4)
```

	movieId	title	genres
0	1	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy
1	2	Jumanji (1995)	Adventure|Children|Fantasy
2	3	Grumpier Old Men (1995)	Comedy|Romance
3	4	Waiting to Exhale (1995)	Comedy|Drama|Romance
4	5	Father of the Bride Part II (1995)	Comedy

	movieId	title	genres	genres_clean
0	1	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure|Children|Fantasy	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy|Romance	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy|Drama|Romance	Comedy Drama Romance
4	5	Father of the Bride Part II (1995)	Comedy	Comedy

Content Based Filtering:

```
recommend_similar_movies("Toy Story (1995)", 5)
```

	title	genres
1706	Antz (1998)	Adventure|Animation|Children|Comedy|Fantasy
2355	Toy Story 2 (1999)	Adventure|Animation|Children|Comedy|Fantasy
2809	Adventures of Rocky and Bullwinkle, The (2000)	Adventure|Animation|Children|Comedy|Fantasy
3000	Emperor's New Groove, The (2000)	Adventure|Animation|Children|Comedy|Fantasy
3568	Monsters, Inc. (2001)	Adventure|Animation|Children|Comedy|Fantasy

User-profile based Filtering

	title	genres
8597	Dragonheart 2: A New Beginning (2000)	Action|Adventure|Comedy|Drama|Fantasy|Thriller
6570	Hunting Party, The (2007)	Action|Adventure|Comedy|Drama|Thriller
4681	The Great Train Robbery (1978)	Action|Adventure|Comedy|Crime|Drama
4005	Flashback (1990)	Action|Adventure|Comedy|Crime|Drama
5471	Diamond Arm, The (Brilliantovaya ruka) (1968)	Action|Adventure|Comedy|Crime|Thriller

Collaborative Filtering:

	title	genres
0	Spawn (1997)	Action|Adventure|Sci-Fi|Thriller
1	Stripes (1981)	Comedy|War
2	Pirates of the Caribbean: The Curse of the Bla...	Action|Adventure|Comedy|Fantasy
3	Tremors (1990)	Comedy|Horror|Sci-Fi
4	Ginger Snaps (2000)	Drama|Horror|Thriller

Matrix Factorization:

	title	genres
507	Terminator 2: Judgment Day (1991)	Action|Sci-Fi
902	Aliens (1986)	Action|Adventure|Horror|Sci-Fi
2078	Sixth Sense, The (1999)	Drama|Horror|Mystery
1445	Breakfast Club, The (1985)	Comedy|Drama
793	Die Hard (1988)	Action|Crime|Thriller

Hybrid Model:

	title	genres
8597	Dragonheart 2: A New Beginning (2000)	Action|Adventure|Comedy|Drama|Fantasy|Thriller
6570	Hunting Party, The (2007)	Action|Adventure|Comedy|Drama|Thriller
4681	The Great Train Robbery (1978)	Action|Adventure|Comedy|Crime|Drama
4005	Flashback (1990)	Action|Adventure|Comedy|Crime|Drama
5471	Diamond Arm, The (Brilliantovaya ruka) (1968)	Action|Adventure|Comedy|Crime|Thriller

Evaluation Output:

```
User 1
Content -> Precision: 0.0 Recall: 0.0
Relevant items: 46
Sample Recommended IDs: [546, 1429, 2835, 4270, 4818]

User 2
Content -> Precision: 0.0 Recall: 0.0
Relevant items: 5
Sample Recommended IDs: [20, 145, 1432, 1473, 2000]

User 3
Content -> Precision: 0.0 Recall: 0.0
Relevant items: 3
Sample Recommended IDs: [611, 1037, 1200, 1320, 2364]

User 4
Content -> Precision: 0.0 Recall: 0.0
Relevant items: 33
Sample Recommended IDs: [72, 3045, 3051, 3067, 3071]

User 5
Content -> Precision: 0.0 Recall: 0.0
Relevant items: 9
Sample Recommended IDs: [145, 1034, 1912, 3266, 3893]
```